

# GenSec Industries: Requirement Specification document



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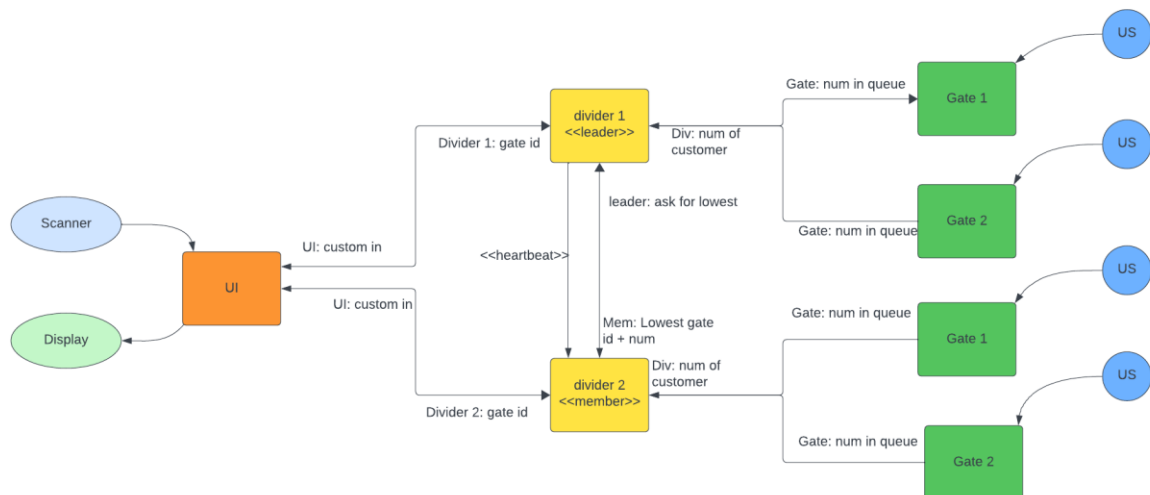
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## 1. Introduction

The focus of this document is to bring together all the requirements that the occupancy management system needs to comply with. These include the user requirements, the functional requirements, and the non-functional requirements. Some requirements were given to the team by the stakeholders. Others were created to give the system a better functionality.

The context diagram of the system is provided below.



**Figure 1:** Context Diagram

## 2. Use Cases

The following use cases were defined for the system. These use cases will be applied for the user requirements in the next section.

Below you will find a brief description of each actor and who they are and what they do.

|    | Actor          | Description (who/what is)                                                                                                                                                                                                                                  |
|----|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. | Passenger      | A <b>passenger</b> is a person who has bought a plane ticket and must pass through the system to get to their correct gate past the security check area. (Primary users of the system).                                                                    |
| 2. | Security Guard | A <b>security guard</b> is a person that must check passengers and their belongings to ensure that they are safe for travel for both them and other around them. Security guards have access to the emergency button that can be triggered in such a case. |

**Note:** The basic flow is what happens in between the pre-condition and the post-condition.

|                               |                                                                                                                                                                                                     |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID:</b><br>UC_001 | <b>Use Case:</b> Choosing entry line                                                                                                                                                                |
| <b>Description:</b>           | An incoming passenger follows the guide on the UI display to choose an entry queue which has low or decent traffic flow.                                                                            |
| <b>Pre-Condition:</b>         | The scanner system is in scanning mode.                                                                                                                                                             |
| <b>Basic Flow</b>             | The passenger scans their ticket on the scanner.<br>The passenger will see the information that is displayed on the screen.<br><br>The passenger follows the guide to go to the correct queue line. |
| <b>Post-Condition:</b>        | The scanner system is back into scanning mode.                                                                                                                                                      |
| <b>Actor:</b>                 | passenger                                                                                                                                                                                           |

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID:</b><br>UC_002 | <b>Use Case:</b> Entering the Security-Check area                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Description:</b>           | A passenger in an entry queue enters the Security-Check area to do a luggage checking procedure before entering the Duty-Free area.                                                                                                                                                                                                                                                                                                     |
| <b>Pre-Condition:</b>         | The divider system is in IDLE mode.                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Basic Flow</b>             | The entry gate is closed, and the number of occupancies is not at maximum.<br><br>The passenger waits in the queue to scan their ticket.<br><br>Once the passenger has scanned their ticket, the divider system will find the gate with the least amount of traffic and assign relays this information back to the passenger.<br><br>The entry gate will open.<br><br>The passenger passes through to get into the security check area. |
| <b>Post-Condition:</b>        | The divider system is back into IDLE mode.                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Actor:</b>                 | passenger                                                                                                                                                                                                                                                                                                                                                                                                                               |

|                               |                                                                                                                                                                                                                                                            |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID:</b><br>UC_003 | <b>Use Case:</b> Entering the Duty-Free area                                                                                                                                                                                                               |
| <b>Description:</b>           | A passenger in the Security-Check Area enters the Duty-Free area after finishing the luggage checking procedure.                                                                                                                                           |
| <b>Pre-Condition:</b>         | The duty-free entrance is closed.                                                                                                                                                                                                                          |
| <b>Basic Flow</b>             | The passenger has just finished their security check. The passenger has collected all their belongings, and they are ready to pass from the security-check room to the duty-free room. After they pass the two sensors, they can enter the duty-free room. |
| <b>Post-Condition:</b>        | The Duty-Free entrance is open.                                                                                                                                                                                                                            |

|               |           |
|---------------|-----------|
| <b>Actor:</b> | passenger |
|---------------|-----------|

|                               |                                                                                                                                                                                                                                                                                            |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID:</b><br>UC_004 | <b>Use Case:</b> Exiting the Security-Check area on Emergency                                                                                                                                                                                                                              |
| <b>Description:</b>           | An airport security guard triggers the emergency protocol somewhere to inform people in the airport to leave the building.                                                                                                                                                                 |
| <b>Pre-Condition:</b>         | <ul style="list-style-type: none"> <li>Alarm trigger is off.</li> <li>There are people inside the Security-Check.</li> <li>Emergency Exits from the Duty-Free area is closed.</li> </ul>                                                                                                   |
| <b>Basic Flow</b>             | The happy flow happens where people are passing through security check to get into the duty-free area. A security guard triggers the alarm that is available on each gate. The alarm goes off. All the passengers must exit the security check area by going back to where they came from. |
| <b>Post-Condition:</b>        | <ul style="list-style-type: none"> <li>Alarm trigger is on.</li> <li>Passengers are leaving the security check area.</li> <li>Security guards leave the security check area after all passengers have been evacuated.</li> </ul>                                                           |
| <b>Actor:</b>                 | passengers & security guards                                                                                                                                                                                                                                                               |

|                               |                                                                                                                                                                                                                       |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID:</b><br>UC_005 | <b>Use Case:</b> Activating an additional divider if the system gets too busy                                                                                                                                         |
| <b>Description:</b>           | When there is too much traffic in both gates 1 and 2 of the currently active divider, then the second divider with gates 3 and 4 will activate and be active for use so that people can be appointed to gates 3 or 4. |
| <b>Pre-Condition:</b>         | <ul style="list-style-type: none"> <li>Divider 1 with gates 1 and 2 are active</li> <li>Divider 2 with gates 3 and 4 are inactive</li> </ul>                                                                          |
| <b>Basic Flow</b>             |                                                                                                                                                                                                                       |
| <b>Post-Condition:</b>        | <ul style="list-style-type: none"> <li>Divider 1 with gates 1 and 2 are active</li> <li>Divider 2 with gates 3 and 4 are active</li> </ul>                                                                            |
| <b>Actor:</b>                 | passengers                                                                                                                                                                                                            |

### 3. Requirements

*In this section, all requirements relating to the use cases above are listed.*

#### 3.1. User Requirements

*User requirements are typically written when discussing the use cases for a project. [1] Below you will find the user requirements for the occupancy management system.*

| User Requirements ID | Description                                                                              | Use Cases ID |
|----------------------|------------------------------------------------------------------------------------------|--------------|
| UR_01                | Passengers should be able to see the UI display on the "Divider" system.                 | UC_001       |
| UR_02                | Passengers should use the UI to be guided to their security lane for the security check. | UC_001       |

|              |                                                                                            |        |
|--------------|--------------------------------------------------------------------------------------------|--------|
| <b>UR_03</b> | Passengers and their luggage need to move together throughout the process.                 | UC_003 |
| <b>UR_04</b> | Passengers should be able to follow the safety rules of the airport during emergencies.    | UC_004 |
| <b>UR_05</b> | Security guards should have access to activate or deactivate the emergency alarm manually. | UC_004 |

## 2.2 Functional requirements

*Functional requirements define what a product must do, what its features and functions are. [2]*  
*Below you will find the functional requirements for the occupancy management system.*

| <b>Functional Requirements ID</b> | <b>Description</b>                                                                                                                                                                                                              | <b>Use Cases ID</b> |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>FR_01</b>                      | The “Divider” system must be able to scan the passenger’s ticket.                                                                                                                                                               | UC_001              |
| <b>FR_02</b>                      | “Divider” system must be able to display the entry queue for the user.                                                                                                                                                          | UC_001              |
| <b>FR_03</b>                      | “Divider” system must be able to decide the entry traffic, based on communication between all active gates.                                                                                                                     | UC_001              |
| <b>FR_04</b>                      | Gate system must be able to communicate with each other and to the “Divider” system.                                                                                                                                            | UC_002              |
| <b>FR_05</b>                      | Gate system must be able to detect its maximum queue to control the traffic flow of passengers.                                                                                                                                 | UC_001              |
| <b>FR_06</b>                      | “Divider” system must be able to communicate with another “Divider” system.                                                                                                                                                     | UC_002              |
| <b>FR_07</b>                      | The occupancy management system’s data must be transferred to the database so a maintainer can download it to check the system if needed. It can also serve to display on a UI to be seen by an employee high in command. (OOS) | UC_002<br>UC_003    |
| <b>FR_08</b>                      | The system should have access to read the alarm trigger of the airport. (OOS)                                                                                                                                                   | UC_004              |
| <b>FR_09</b>                      | The alarm trigger should be accessible so that a security guard can trigger an emergency case.                                                                                                                                  | UC_004              |

## 2.3 Non-functional requirements

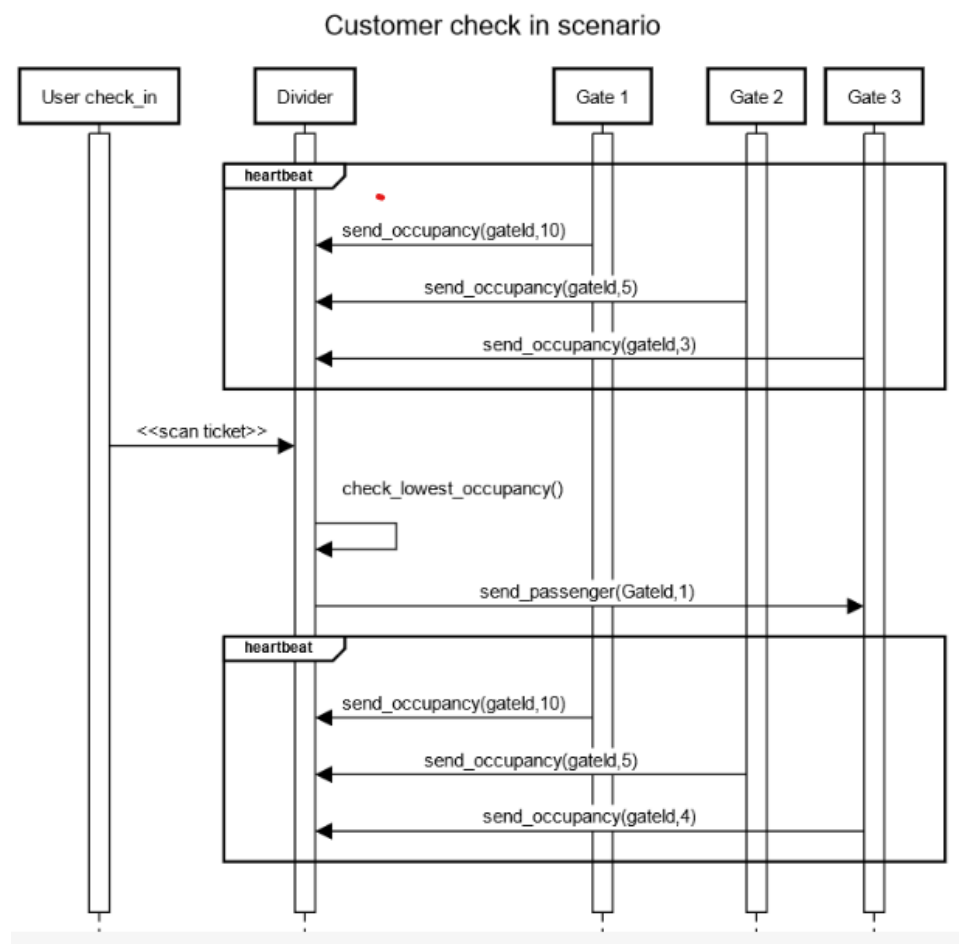
*Non-functional requirements describe the general properties of a system. They are also known as quality attributes. [2]* Below you will find the non-functional requirements for the occupancy management system.

| <b>Non-Functional Requirements ID</b> | <b>Name</b> | <b>Description</b>                                                                                            | <b>Use Cases ID</b>        |
|---------------------------------------|-------------|---------------------------------------------------------------------------------------------------------------|----------------------------|
| <b>NFR_01</b>                         | Performance | Response time of communication between the Gate Systems to the “Divider” system and each other must be quick. | UC_001<br>UC_002<br>UC_003 |

|               |                 |                                                                                                                                      |                  |
|---------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <b>NFR_02</b> | Reliability     | If one gate disconnected, the system should communicate to decide a solution                                                         | UC_001           |
| <b>NFR_03</b> | Maintainability | The system must notify the user if a malfunction happens, and the system must be extendable and upgradable for future possibilities. | UC_001<br>UC_002 |
| <b>NFR_04</b> | Error Handling  | The system must display a clear and informative message of error at a malfunction in case of access denial.                          | UC_001<br>UC_002 |

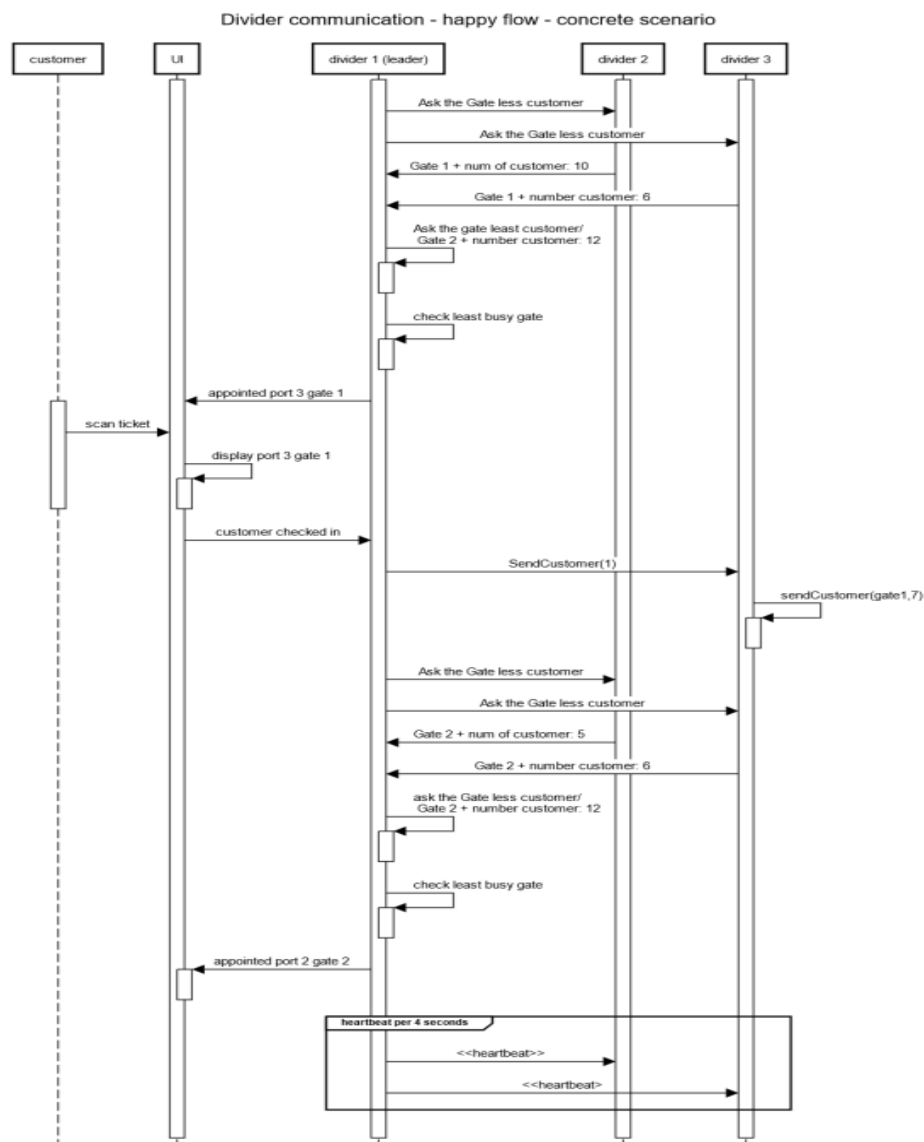
## 2.4 Concrete scenario Skeleton

The concrete scenario is defined so that there is a good foundation to work upon. The concrete scenario explains use case: UC\_002 better by providing this with a sequence diagram. Below you will find the concrete scenario. NOTE: this version has the heartbeat method implemented so that the Divider can keep track of all the "alive" gates.



## 2.5 Concrete scenario version 2

In this version of the concrete scenario, the previous concrete scenario is taken and improved upon. A brief description will be given on what is happening along with a sequence diagram to show the functionality of each component.





## 4. References

- [1] - *User requirement*. User Requirement - an overview | ScienceDirect Topics. (n.d.).  
<https://www.sciencedirect.com/topics/engineering/user-requirement>
- [2] - Editor. (2019, October 18). *Functional and nonfunctional requirements: Specification and types*. AltexSoft. <https://www.altexsoft.com/blog/business/functional-and-non-functional-requirements-specification-and-types/>